

Prajwal Chikkur

Software Engineer | Backend Engineer | Generative AI Engineer

Bangalore, Karnataka, India · chikkurprajwal077@gmail.com · +91-7204764005 · linkedin.com/in/prajwal-chikkur

SUMMARY

Software Engineer with 1+ year of professional experience building backend systems, Generative AI applications, and automation platforms using Python, FastAPI, PostgreSQL, Redis, Celery, and AWS. Experience spans scalable workflows, asynchronous processing systems, REST APIs, production AI applications, observability systems, and performance optimizations.

EXPERIENCE

SeedlingLabs

December 2025 - Present

Software Engineer I

Bangalore, Karnataka

Orchard — AI-Powered Testing Automation

Python · PostgreSQL · Playwright · BrowserUse · Celery · Redis · GenAI

Re-architected an AI-driven UI and API testing platform using Python, PostgreSQL, browser automation, and asynchronous processing, accelerating automated test creation by 15x.

- Developed backend workflows that translate natural-language requirements into executable browser and API testing actions.
- Designed scalable execution infrastructure using Redis for session and state management and Celery for asynchronous task processing.
- Enabled concurrent execution of long-running testing workloads.
- Reduced memory consumption by 80% through tab-level browser reuse and optimized resource management.
- Implemented self-healing tests, intelligent assertions, automated reporting, and failure-handling workflows.
- Integrated execution tracing and observability using Laminar to monitor production workflows, identify failures, and improve system performance.

Sprout — AI-Powered School Operating Platform

Python · FastAPI · PostgreSQL · Redis · Celery · AWS Bedrock · Terraform · GenAI

Built production backend services for an AI-powered school platform supporting automated generation of lesson plans, worksheets, question papers, and educational content.

- Built production backend services using Python and FastAPI.
- Designed asynchronous processing pipelines using Celery and Redis for large-scale handwritten answer evaluation.
- Engineered a multi-provider model routing layer across AWS Bedrock, working with Claude, Gemini, and Sarvam, selecting models based on workload requirements.
- Tracked token usage, latency, and cost, and implemented model and workload optimization strategies.
- Designed deterministic diagram-generation workflows using RDKit, Graphviz, and CairoSVG.
- Implemented AI and application observability using OpenTelemetry, Laminar, and Prometheus/Grafana.

PROJECTS

ExamGPT — RAG-Based AI Exam Assistant

Python · FastAPI · PostgreSQL · RAG · Gemini

Built a document-based question-answering system that processes uploaded academic PDFs and generates contextual answers using Retrieval-Augmented Generation. Features PDF document ingestion, text processing, semantic retrieval, and contextual prompting.

Intelligent Voice Assistant

Python · FastAPI · LLMs · REST APIs

Developed a multi-platform AI assistant with voice interaction, REST API integrations, conversational workflows, and tool-based task execution, built on a modular backend architecture.

RAG & MCP Based Employee Management System

Python · RAG · MCP · PostgreSQL

Developed an employee management application combining relational data management with retrieval-augmented AI capabilities, enterprise document retrieval, and contextual AI responses for enterprise queries.

TECHNICAL SKILLS

Programming

Python, C++, C, Java, JavaScript, C#

Backend & APIs

Python, FastAPI, REST APIs, PostgreSQL, SQLite, Redis, Celery, Asynchronous Processing

Systems & Infrastructure

Docker, AWS, Terraform, CI/CD, Git, GitHub, Scalable Systems, Concurrent Processing, Distributed Workflows

Generative AI

Generative AI, LLMs, RAG, AWS Bedrock, Claude, Gemini, OpenAI APIs, Prompt Engineering, Model Routing, LLM Observability

Testing & Development

Playwright, Browser Automation, API Testing, Automated Testing, React, Next.js, React Native

EDUCATION

Bachelor of Computer Applications (BCA)

9.07 / 10

Sri Vidyaniketan Degree College, Gangavathi · 2023 – 2026

CERTIFICATIONS

Generative AI — Udemy

Prompt Engineering — Udemy

Cybersecurity Analyst Job Simulation — Forage (TATA)